SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Automated Attendance System

1. Introduction

1.1 Purpose

The purpose of the Automated Attendance System is to automate the process of recording and managing student attendance. The system reduces manual work, saves time, minimizes errors, and provides accurate attendance records for students and faculty.

1.2 Scope

The system allows faculty to:

Register students and faculty.

Create and manage classes/subjects.

Record student attendance automatically.

View daily and monthly attendance.

Generate attendance reports.

Identify students with low attendance.

Maintain attendance records securely.

Students can:

Login to the system.

View their attendance percentage.

Check subject-wise attendance.

View attendance history.

1.3 Definitions

Term

Meaning

SRS

Software Requirements Specification

Admin

Person who manages the system

Faculty

Staff member who records attendance

Student

User whose attendance is recorded

Attendance

Record of student’s presence or absence

Database

Storage used to maintain system data

2. Overall Description

2.1 Product Perspective

The Automated Attendance System is a web/mobile-based application connected to a database.

Basic flow:

Student / Faculty

↓

Login

↓

Automated Attendance System

↓

Attendance

Processing

↓

Database

↓

Reports / Dashboard

2.2 User Classes

Admin

Manage students

Manage faculty

Manage subjects

Manage classes

View reports

Faculty

Login

Start attendance

View attendance

Edit attendance if authorized

Generate reports

Student

Login

View attendance

View attendance percentage

View attendance history

3. Functional Requirements

FR1 – User Login

The system shall provide secure login for Admin, Faculty and Students.

FR2 – Student Management

Admin shall be able to:

Add students

Update student details

Delete student records

View student information

FR3 – Faculty Management

Admin shall be able to add, update and manage faculty information.

FR4 – Subject Management

Admin shall be able to create and manage subjects and assign them to faculty.

FR5 – Automated Attendance

The system shall automatically record attendance based on the configured attendance mechanism.

For example, depending on the project implementation, attendance can be recorded using:

QR code

Face recognition

RFID

Biometric authentication

FR6 – Attendance Verification

The system shall verify the student before marking attendance and prevent duplicate attendance for the same session.

FR7 – Attendance Records

The system shall store:

Student ID

Student name

Subject

Date

Time

Attendance status

FR8 – Attendance Calculation

The system shall automatically calculate attendance percentage.

Formula:

Attendance % =

(Present Classes / Total Classes) × 100

FR9 – Attendance Reports

Faculty/Admin shall be able to generate:

Daily attendance report

Subject-wise report

Student-wise report

Monthly attendance report

FR10 – Low Attendance Alert

The system shall identify students whose attendance falls below the required percentage and display an alert.

FR11 – Student Dashboard

Students shall be able to view their attendance percentage and attendance history.

FR12 – Data Management

The system shall securely store attendance and user information in the database.

4. Non-Functional Requirements

4.1 Performance

The system should process attendance quickly.

Attendance records should be displayed without significant delay.

The system should support multiple users.

4.2 Security

User login should be authenticated.

Only authorized users can modify attendance.

Student data should be protected.

4.3 Reliability

Attendance records should not be lost.

The system should maintain consistent attendance data.

4.4 Usability

The interface should be simple and user-friendly.

Faculty and students should easily understand the dashboard.

4.5 Availability

The system should be available whenever attendance needs to be recorded or viewed

5. Hardware Requirements

Minimum Requirements

Computer/Laptop or Smartphone

Processor: Intel i3 or equivalent

RAM: 4 GB or above

Storage: 10 GB free space

Internet connection

Additional hardware depends on the attendance method, such as a camera for face recognition or an RFID reader for RFID-based attendance.

6. Software Requirements

Front End

HTML

CSS

JavaScript

Back End

Java / Python / Node.js

Database

MySQL / Firebase

Development Tools

VS Code / Eclipse / IntelliJ IDEA

Web browser

Note: If your project is specifically a Flutter app, front end can be Flutter/Dart and backend can be Firebase or Node.js.

7. Database Requirements

Important tables can include:

Student Table

Student_ID

Name

Department

Year

Section

Email

Password

Faculty Table

Faculty_ID

Name

Department

Email

Password

Subject Table

Subject_ID

Subject_Name

Faculty_ID

Department

Attendance Table

Attendance_ID

Student_ID

Subject_ID

Date

8. Future Enhancements

AI-based face recognition.

Mobile application.

Parent notification through SMS/email.

Cloud-based attendance management.

Attendance analytics and dashboards.

Integration with college ERP/LMS.

Automatic timetable-based attendance.

9. Conclusion

The Automated Attendance System provides an efficient way to record, manage and monitor student attendance. By automating attendance collection and report generation, the system reduces manual effort.